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BRIEF REPORT

Another Gorilla (*Gorilla gorilla gorilla*) Recognizes Himself in a Mirror

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Various attempts have been made to explain why gorillas (*Gorilla gorilla gorilla*) find it difficult to recognize their mirror image. One of the most oft-cited reasons is aversion to eye contact, which stops gorillas from looking into a mirror and thus prevents them from carrying out a suitable exploration that could lead to self-recognition. In the experimental design used here the subject was first habituated both to observers and to the mirror as an object before being exposed to the latter. The study was performed with a single subject who was well adapted to captivity and exhibited no aberrant behavior or signs of stress. The results revealed that the subject had no aversion to eye contact. He showed considerable interest in the mirror and appeared relaxed when faced with his image. He gave a positive response to the mark test. Am. J. Primatol. 69:576–583, 2007. © 2006 Wiley-Liss, Inc.

Key words: gorilla; self-recognition; mark test; habituation

INTRODUCTION

In the early 1970 s G.G. Gallup, Jr., was the first to use an innovative and systematic method to evaluate the capacity of subjects with no language ability to recognize themselves in mirrors. This method, known as the mark test [Gallup, 1970], consists of applying a paint mark to an area of the body that is only visible with the use of a mirror.

To date, evidence of self-recognition has been found in chimpanzees [Povinelli et al., 1997], bonobos [Hyatt & Hopkins, 1994], and some orangutans [Lethmate & Dücker, 1973]. However, the results from gorillas have proven controversial: of 23 gorillas studied [Swartz et al., 1999], six showed self-directed behavior guided by their mirror image, and only five produced a possible positive result on the mark test [Patterson & Cohn, 1994; Parker, 1994; Swartz & Evans, 1994].

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Various attempts have been made to explain this apparent failure in gorillas [Povinelli, 1987; Gallup et al., 1980; Parker, 1994; Patterson & Cohn, 1994].

Some of these hypotheses have been tested; for example, Shillito et al. [1999] recorded the behavior of gorillas in front of a mirror in the absence of observers, in order to avoid a possible human influence on their behaviors, but the results were not positive. They then used two mirrors placed at an angle of 60° in order to avoid a direct gaze, but this failed to improve the results. With respect to gorillas' interest in exploring their own body, Suarez and Gallup [1981] marked subjects on the wrist and observed that they showed an interest in the marks; however, the animals ignored the mark that was not visible without the help of the mirror.

MATERIALS AND METHODS

The subject of the present study (Xebo, 17 years old) was an adult male of the species *Gorilla gorilla gorilla* from Barcelona Zoo. He was raised by his own mother in a social group. Currently he lives with two adult females, is well adapted to captivity, exhibits no anomalous behavior, and is a fairly calm animal. During the observations Xebo was alone in an inside enclosure, a cage 4 m wide by 3 m long. The mirror was placed outside the cage, 0.2 m away from the bars. A Perspex mirror measuring 150 × 100 cm was used. Data were collected with a Panasonic RQ-L30 recorder. Moreover, everything the subject did in front of the uncovered mirror was recorded with a Canon DM-XM1 digital video camera. To mark the subject, an odorless, non-irritating yellow makeup stick designed for children was used.

The procedure was divided into four phases: 1) *Habituation and ethogram*: The goal was to habituate the subject to the observer, and to draw up and test the ethogram (total hours of observation = 40). (The ethogram is given in Appendix A.) 2) *Presentation of the covered mirror*: The goal was to habituate the subject to the presence of the mirror. Sampling was focal and concerned behavior; recording was continuous. For *staring* behavior, time was considered as well as frequency (total hr of observation = 16). 3) *Presentation of the uncovered mirror*: Both sampling and systematic recording were performed as in the previous stage (total hr of observation = 28 hr; 16 sessions lasting 1 hr 45 min each). 4) *Mark test*: One week after the last session of the uncovered-mirror phase, the subject (after a session of training using water) was marked with paint on the forehead without the use of anesthetic by his usual keeper. Before the mirror was introduced, a 30-min baseline period was allowed to elapse so that we could see how many times he spontaneously touched the marked area. The mirror was then introduced and the subject was observed for 45 min. One week after the test mark was applied, a control procedure was also performed in which a mark of the same color was painted on the subject's arm, hand, and abdomen.

The same observer was used for each phase of the study. In phases 3 and 4 a second person video-recorded the sessions. Intraobserver reliability was tested using video-recorded material, and agreement was high (Cohen's $\kappa = 0.899$).

RESULTS

From the outset Xebo appeared relaxed when faced with his mirror image. He exhibited no agonistic behavior and actually showed considerable interest in the mirror. Data from the first session indicated 64 episodes of *staring*, which lasted a total of 910 sec, and 57 *glances*. The overall time dedicated to looking at the mirror across all exposure sessions was quite high (total = 13,444 sec).

At times the subject *stared* at the mirror without doing anything else, while at other times the *staring* was accompanied by a *self-referred* action. The most frequent behaviors performed while looking at the mirror were *scratching* (92) and *pulling face* (43). The *pulling-face* behaviors consisted of opening his mouth, sticking his tongue out, and extending it toward his upper lip as if he wanted to touch his nose with the tip of his tongue, *staring at himself* in the mirror, and pulling his upper lip upwards with his right hand. The subject touched his forehead on five occasions during the uncovered-mirror phase, without looking at the mirror.

On the fifth day there was a change in the subject's behavior and an increase in *self-referred* behaviors, some of which were exhibited in body postures that until that day had not been observed. For example, he adopted a supine position with his head facing the mirror, at an angle that allowed him to see his reflection while carrying out a *self-referred* action (mainly *scratching*).

Mark Test

During the 30 min of baseline study there was no *self-directed* action targeting the marked area. The uncovered mirror was presented for 45 min. He *stared* at the mirror on 27 occasions (equivalent to a total of 939 sec looking at the mirror), and performed 15 *glances*. The amount of time dedicated to *staring* at the mirror increased with respect to the last day of exposure. The subject looked at the mirror in a relaxed way. During the first 30 min he *stared* at the mirror for a long time, sometimes for more than 1 min, and most of these behaviors were unaccompanied by any other kind of action. He sometimes turned his head slightly from right to left successively, as if he wanted to see his profile. During the last 15 min, having *stared* at the mirror on four occasions for 60–90 sec, he moved away from the mirror and *touched* the marked area 15 times without looking at himself, *sniffed* his hand seven times, and *looked* at it five times. First he *touched* his forehead, and then sometimes *touched* it again or *sniffed* or *looked* at his hand before repeating these actions. After exhibiting this behavior at the frequencies indicated, and now looking at the mirror, he *rubbed* the forehead on three occasions and *touched* it a further three times; he also *sniffed* his hand twice after he *touched* his forehead. The total time spent carrying out these actions while looking at the mirror was 350 sec. He stopped performing these actions when the mark disappeared from his body. In the control experiment conducted to observe the degree of interest shown by the subject in the marks on his body, patches of color were painted on his arm, hand, and abdomen, which he immediately wiped off.

To compare the results obtained with the uncovered mirror and the uncovered mirror with the mark (Table I), confidence intervals (CI = 95%) were determined. For the self-directed behavior with the uncovered mirror, there were 38.86–23.06 behaviors/hr ($\overline{X} = 30.96$; Standard deviation [S] = 16.1; CI 95% = $\pm$7.90), while in the mark test Xebo performed 61.33 self-directed behaviors per hour. For the self-referred behavior with the uncovered mirror, the CI was 8.54–2.46 behaviors/hr ($\overline{X} = 5.5$; S = 6.2; CI 95% = $\pm$3.04), while in the mark test Xebo performed 10.66 self-referred behaviors per hour. The CI for duration of staring at the uncovered mirror was 0.282–0.152 hr ($\overline{X} = 0.217$; S = 0.133; IC 95% = $\pm$0.065), and in the mark test Xebo stared for 0.260 hr. Therefore, for the self-directed and self-referred behavioral categories the difference was significant ($\alpha = .05$), while the duration of staring fell inside the CI.

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TABLE I. Frequency and Percentage of the Ethogram Categories in Covered Mirror, Uncovered Mirror, and Uncovered Mirror With Mark

	Phases					
	Covered		Uncovered		Uncovered with mark	
Categories	Frequency	%	Frequency	%	Frequency	%
Agonistic	1	2%	0	0%	0	0%
Exploratory	72	12.9%	349	9.4%	0	0%
Contingent	0	0%	0	0%	0	0%
Self-direct	89	15.9%	867	23.3%	46	45.1%
Self-referred	0	0%	154	4.1%	8	7.8%
Guided	0	0%	0	0%	6	5,9%
Lookings						
Glance	347	62%	1260	33,9%	15	14.7%
Stare	51	9,1%	1085	29,2%	27	26,5%
Total	560	100%	3751	100%	102	100%

DISCUSSION

The results illustrate the high degree of curiosity shown by the subject toward the mirror, as well as the apparent lack of any aversion to direct eye contact, since there were numerous occasions on which he stared into the mirror, sometimes for lengthy periods. The lack of aggression shown toward his own mirror image was evident from the outset; moreover, at no point did we notice the acrid smell that emanates from adult male gorillas in moments of excitation or tension, which is an indirect indicator of the lack of stress experienced by Xebo in front of the mirror. It is possible that this relaxed attitude favored the exploration of his image and helped him understand that he was not face to face with an unknown adult male gorilla, as occurred in the case of Pogo and Bwana described by Parker [1994]. It therefore has to be asked whether Xebo's response on the mark test fulfills the criteria for consideration as a positive response. According to Gallup [1970], the result of the mark test should be considered positive when the subject touches and cleans the marked area while looking into the mirror. But what happens in the case of Xebo and other gorillas, such as Michael [Patterson & Cohn, 1994] and Bwana [Parker, 1994], that touch and clean the marked area while looking away from the mirror? A detailed analysis of this situation suggests the following:

First, in the case of Xebo we see that after he was marked he did not touch that area–neither immediately afterwards nor during the 30-min baseline period. This indicates that he was not aware of having been marked, and that he received no olfactory or tactile information that suggested there was anything unusual on his body. In contrast, in the case of Bwana [Parker, 1994] the subject did show awareness at the time he was marked. With respect to the behavior *touching himself* in the marked area, during the 28 hr of exposure to the uncovered mirror, prior to the mark test, the subject was observed to *touch himself* in this area on five occasions. During the 45 min of exposure to the mirror on the day the mark was applied, he *touched himself* in this area 21 times. The first 15 times he did so

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without looking directly into the mirror, but this was immediately followed by prolonged staring (at times for over 1 min) at his mirror image; the last six times included looking at himself in the mirror.

In addition to the increased frequency with which he touched this area, another important aspect is the behavior shown immediately after he *touched himself*: this behavior was *smelling himself* and *looking at* the fingers with which he had *touched* the mark. Gallup [1970] viewed the visual and olfactory inspection of the hands after touching the mark as particularly noteworthy evidence of self-recognition. This behavioral sequence only makes sense if the subject is aware that the mark is on his own body [Gallup, 1975], and was only observed during the mark test after he *looked* at himself in the mirror. A similar behavior has also been observed in some chimpanzees [Gallup, 1970] and gorillas [Patterson & Cohn, 1994; Swartz & Evans, 1994] who produced a positive result on the mark test. The fact that Xebo wiped the mark clean just after he *looked* at himself in the mirror implies that 1) he recognized himself, since he was aware that the mirror image was his own and that therefore the mark was on his own body rather than on the mirror; 2) the subject had a good knowledge of his body schema and a good short-term memory, since at first he did not use the mirror as a tool to guide himself; and 3) the subject used the mirror as a tool, and did so to check that the mark has completely disappeared from his body, since after he cleaned himself without looking at himself he moved toward the mirror again and cleaned himself until the mark disappeared, and he did not touch the area afterwards. Similar behavior was observed in a male dolphin by Marten and Psarakos [1994].

Why did Xebo, unlike so many of the gorillas tested, pass the mark test? He is not an acculturated gorilla like Koko [Patterson & Cohn, 1994]; on the contrary, he was raised by his own mother and grew up in a social group. Unlike other gorillas in captivity, he shows no anomalous behavior or signs of stress when faced with observers or an experimental situation. Gorilla populations in captivity are known to suffer frequently from handling and rearing problems that affect their behavior [Mariner & Drickamer, 1994]. Xebo is a sexually active, reproductive male that has good relationships with the females of his group. It is likely that his good adaptation to captivity, his social group, and species-suitable rearing conditions have combined to produce a state of psychological health that favors the development and expression of the capacity for self-recognition via the mark test.

The evidence of self-recognition in gorillas, whether through the mark test or as a result of behavior guided by their mirror image, indicates that gorillas do indeed show the capacity for self-recognition. Thus, the present results do not support the hypothesis of Gallup and Povinelli, which states that the genus *Gorilla* has lost the capacity for self-recognition. This capacity would be present in those subjects, such as Xebo, whose social environment has favored an adequate physical and psychological development, and would show itself provided that the experimental conditions include a prior period of habituation and respect the particular characteristics of the species.

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APPENDIX A: ETHOGRAM

Agonistic Behavior With Respect to the Mirror

Moving away: Starting from a distance of less than 1 m from the mirror, the subject begins to move away such that his reflection can no longer be seen.

Display: A set of intimidating actions that are presented at a moment of tension, such as making a noise by banging parts of the installation, beating the chest, running across the installation, and hair standing on end. All of these actions are accompanied by the release of an intense acrid odor that is easily recognizable.

Striking: The subject strikes the bars with one of his extremities or an object, or shakes the bars.

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Throwing: The subject throws an object (straw, food, excrement, etc.) with force against the bars (includes spitting).

Exploratory Behavior With Respect to the Mirror

Moving closer: The subject moves closer to the mirror while looking at it.

Giving: Giving an object, trying to pass it through the bars, or behavior associated with asking for things, such as stretching a hand out toward the mirror or blowing kisses.

Attempt to touch the mirror: Pushing fingers through the bars in an attempt to reach the surface of the mirror.

Self-Directed Behavior, Self-Referred Behavior, and Behavior Guided by the Mirror

The same behaviors appear under these different categories, but their categorization depends on the relation of the behavior to the mirror. When the subject carries out the actions with and on his own body, but without seeing himself reflected in the mirror, his behaviors are *self-directed behaviors*. When the subject carries out the actions with and on his own body while looking in the mirror, but without these actions being guided by the mirror image, his behaviors are *self-referred behaviors*. And when the subject carries out the actions with and on his own body, and the actions are clearly guided by the mirror image (with the mirror being used as a tool that enables the actions to be performed), his behaviors are *behaviors guided by the mirror image.*

Licking himself: Moving the tongue over an area of the body surface.

Looking at himself: Staring at a part of the body without doing anything else.

Picking at an orifice: Introducing a finger into any body cavity (nasal, ocular, auditory, vaginal, or anal) or between the teeth.

Pulling at himself: Grabbing a lock of hair and pulling on it strongly.

Pulling faces: Sticking out his tongue, screwing up his face, etc. In general, any movement of the face muscles that forms part of the species repertoire are characteristic of the subject.

Putting on/taking off: Any external object (including excrement) is placed on or removed from any part of the body.

Rubbing himself: Moving the surface of the fingers repetitively over an area of the body (the intensity of the movements may vary).

Scratching himself: Toe- or fingernails are moved vigorously over an area of the subject's body surface.

Self-grooming: The subject uses his hands, feet, and/or mouth to clean, groom, and wash himself of parasites, dry skin, or particles, moving hair aside and staring at this specific part of the body.

Sniffing himself: Moving the nose very close to a body part, to a distance of less than 2 cm away.

Taking hold of himself: Use of the hand or food to take hold of a part of the body.

Touching himself: Contact between toes/fingers or the whole food/hand and an area of the body surface but without movement; this may be fleeting or prolonged.

Yawn: Opening his mouth; teeth may or may not be shown.

[1]Contingent behavior:

All behaviors exhibited by the subject while he is looking into the mirror that involve repetitive movements of a part of the body.

Looking at the Mirror

Glance: Looking at the mirror for less than 1 sec.
Stare: Looking at the mirror for more than 1 sec.

[1]This ethogram was made from a sample of eight gorillas, including our subject, but Xebo did not show this behaviour.

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